*Editorial*

Transactions of the Institute of Measurement and Control
34(6) 665–667

DOI: 10.1177/0142331211419453
tim.sagepub.com

# New trends on bio-inspired computation

**Zhihua Cui[1,2], Zhongzhi Shi[2] and Janos Abonyi[3]**

Biological systems are self-organizing, tolerant of manufacturing defects and they adapt, rather than being programmed, to their environments. The problems they solve involve the interaction of an organism/system with the real world. Bio-inspired computation is the use of computers to model nature, and simultaneously the study of natural biological systems to improve the usage of computers. It is a major subset of natural computation. Recently, bio-inspired computation, such as evolutionary computation, swarm intelligence bacterial foraging, cultural algorithms, neural networks, fuzzy systems, rough sets, molecular computing and other bionics, is becoming increasingly important in face of the complexity of today's demanding applications.

This special issue on bio-inspired computation is dedicated to the latest work in the theory and applications in this exciting area. Our aim is to provide a useful reference for understanding new trends in bio-inspired computation. After a detailed review process, eight papers were selected to reflect the thematic vision. The contents of these studies are briefly described as follows.

The group search optimizer (GSO) was inspired by animal social searching behaviour. It has been shown that the global search performance of the GSO is competitive to other biologically inspired optimization algorithms. In the paper, 'Analysis of premalignant pancreatic cancer mass spectrometry data for biomarker selection using a group search optimizer', S. He et al. apply a GSO as a feature selection method to mass spectrometry (MS) data analysis for premalignant pancreatic cancer biomarker discovery. After applying a smooth nonlinear energy operator (SNEO) to detect peaks, then a GSO with linear discriminant analysis (LDA) is used to select a parsimonious set of peak windows ((biomarkers) that can distinguish cancer. After selecting a set of biomarkers, a support vector machine (SVM) is then applied to build a classifier to diagnosis premalignant cancer cases. The GSO algorithm was compared with a genetic algorithm (GA), evolution strategies (ES), evolutionary programming (EP) and a particle swarm optimizer (PSO). The results showed that the GSO-based feature selection algorithm is capable of selecting a parsimonious set of biomarkers to achieve better classification performance than other algorithms.

In the paper, 'Viral system algorithm: foundations and comparison between selective and massive infections', Pablo Cortés et al. present a guided and deep introduction to viral systems (VS), a novel bio-inspired methodology based on the natural biological process taking part when the organism has to give a response to an external infection. VS has proven to be very efficient when dealing with problems of high complexity. The paper discusses on the foundations of viral systems, presents the main pseudocodes that need to be implemented and illustrates the methodology application. A comparison between VS and other metaheuristics, as well between different VS approaches, is presented. Finally, trends and new research opportunities are presented for this bio-inspired methodology.

Population-based heuristic optimization methods like differential evolution (DE) depend largely on the generation of the initial population. The initial population not only affects the search for several iterations but often also has an influence on the final solution. The conventional method for generating an initial population is the use of computer-generated pseudorandom numbers, which may not be very effective. In the paper entitled with 'A simplex differential evolution algorithm: development and applications', Musrrat Ali et al. have investigated the potential of generating the initial population by integrating the non-linear simplex method (NSM) of Nelder and Mead with pseudorandom numbers in a DE algorithm. The resulting algorithm named the non-linear simplex differential evolution (NSDE) is tested on a set of 20 benchmark problems with box constraints and two real-life problems. Numerical results show that the proposed scheme for generating the random numbers significantly improves the performance of DE in terms of fitness function value, convergence rate and average CPU time.

In the paper, 'How to design a powerful family of particle swarm optimizers in inverse modelling', Juan Luis Fernández Martínez et al. show how to design a powerful set of particle swarm optimizers for application in inverse modelling.

[1]Complex System and Computational Intelligence Laboratory, Taiyuan University of Science and Technology, Taiyuan, Shanxi, China
[2]The Key Laboratory of Intelligent Information Processing, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China
[3]Department of Process Engineering, University of Veszprém, Hungary

**Corresponding author:**
Zhihua Cui, Complex System and Computational Intelligence Laboratory, Taiyuan University of Science and Technology, Taiyuan, Shanxi, China
Email: Cuizhihua@gmail.com

The design is based on the interpretation of the swarm dynamics as a stochastic damped mass-spring system, the so-called PSO continuous model. Based on this idea, a family of PSO optimizers (GPSO, CC-PSO and CP-PSO) was derived, having different exploitation and exploration capabilities. Their convergence is related to the stability of their first- (mean trajectories) and second-order moments (variance and temporal covariance). Good parameter sets are located inside their first stability regions close to the upper border of their respective second stability regions, where the attraction from the particles oscillation centre is very weak. In this region of weak attraction, both convergence to the global minimum and exploration of the search space are possible. Based on this idea, a particle–cloud algorithm was designed where each particle in the swarm has different inertia (damping) and acceleration (rigidity) constants. The performance of these algorithms for different PSO members using different benchmark functions is tested, and the results show that the cloud algorithms have a very good balance between exploration and exploitation. Also, the cloud design helps to avoid the two main drawbacks of the PSO algorithm: the tuning of the PSO parameters and the clamping of the particles' velocities.

Supply chain planning has been regarded as a key strategic decision-making activity for enterprises under the current business circumstances. From the point of supply chain planning, the important issues are to find suitable and quality partners and to decide upon an appropriate production-distribution plan. In the paper, 'Using analytic network process and turbo particle swarm optimization algorithm for non-balanced supply chain planning considering supplier relationship management', Z.H. Che et al. address the development of a decision methodology for supply chain planning in a multi-echelon non-balanced supply chain system, taking into account such criteria as cost, quality, delivery and the supplier relationship management (SRM), considering quantity discount and capacity constraints. The proposed methodology is based on the analytic network process (ANP) and turbo particle swarm optimization (TPSO), to evaluate partners and to find an optimal supply chain network pattern and production-distribution mode. Finally, to demonstrate the performance of the proposed TPSO algorithm, comparative numerical experiments are performed by TPSO, particle swarm optimization (PSO) and GA. Empirical analysis results demonstrate that TPSO can outperform PSO and GA in non-balanced supply chain planning problems.

In the paper, 'An analytical approach to the similarities between swarm intelligence and artificial neural network', Renbin Xiao et al. focus on swarm intelligence represented by the ant colony system and an artificial neural network simulating the brain working principles, and analyse the similarities between them by the approaches of complexity study. Firstly, the similarities between swarm intelligence and neural network have been analysed qualitatively from system structure and operation mechanisms, and the results show that there exist obvious similarities in overall disciplines between them. Moreover, the mapping relationships between swarm intelligence and artificial neural network at algorithm level have been revealed using a travelling salesman problem (TSP), which illustrates that these two algorithms are also similar from the viewpoint of numeric computing and simulating when they are directly used to solve complex problems. In addition, the swarm intelligence algorithm and the artificial neural network algorithm are adopted to solve an engineering practical problem, which can be transformed into the TSP, respectively. The results show that the performance of these two algorithms still has close similarities. The purpose of the above work is to construct the inherent relationship among different bio-inspired computation methods, which has both important theoretic significance and practical value to reveal the generation and operation mechanisms of human intelligence.

In the paper, 'Economic load dispatch using population-variance harmony search algorithm', B.K. Panigrahi et al. present a novel stochastic optimization approach to solve a constrained economic load dispatch (ELD) problem using harmonic search algorithm (HS). HS is a recently developed derivative-free, meta-heuristic optimization algorithm, which draws inspiration from the musical process of searching for a perfect state of harmony. This work analyses the evolution of the population variance over successive generations in HS and thereby draws some important conclusions regarding the explorative power of HS. The proposed methodology easily takes care of solving non-convex ELD problems, along with different constraints like transmission losses, ramp rate limits of the generators and prohibited operating zones. Simulations were performed over various standard test systems with a different number of generating units and a comparative study is carried out with other existing relevant approaches. The findings affirmed the robustness and proficiency of the proposed methodology over other existing techniques.

In the paper, 'Some techniques for analysing time complexity of evolutionary algorithms', Lixin Ding and Jinghu Yu introduce some techniques for analysis of time complexity of evolutionary algorithms (EAs for brevity) based on finite search space. Markov property and the decomposition of a matrix are employed for the exact analytic expressions of the mean first hitting times when EAs reach the optimal solutions (FHT-OS for brevity). Dynkin's formula, one-step increment analysis and some other probability methods are adopted for the lower and upper bounds of the mean FHT-OS. The theory of the non-negative matrix, including the Perron–Frobenius theorem, Jordan Standard form etc., are applied for the convergence of EAs. In addition, some techniques for time complexity and convergence of general adaptive EAs are also involved in this paper. Finally, the theoretical results obtained in this paper are verified effectively by applying them to the analysis of a typical EA, (1+1)-EA.

The guest editors of this special issue in *Transactions of the Institute of Measurement and Control* would like to thank all authors for submitting their interesting work. We are grateful to the reviewers for their great contributions to this special issue. We would like to express our sincere appreciation to the Editors-in-Chief, Prof. Steve Thompson, Prof. John O. Gray and Prof. Frank L. Lewis, for their advice and support.

This special issue has been supported by National Natural Science Foundation of China under Grant 61003053, the Key

Project of Chinese Ministry of Education under Grant 209021, Shanxi Science Foundation for Young Scientists under Grant 2009021017-2, the National Natural Science Foundation of China under Grant No. 60775035, 60903141 and 60970088, the Major State Basic Research Development Program of China under Grant No. 2007CB311004, the National Science and Technology Support Plan under Grant No. 2006BAC08B06 and the Major Program of National Natural Science Foundation of China under Grant No. 60933004.